

A CLASS OF COMPOUND DISTRIBUTIONS OF THE REPARAMETERISED BIVARIATE GAMMA EXTENSION

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Summary: Steel and Le Roux (1987) introduced a reparameterisation of a bivariate gamma extension that may be used to model a variety of physical situations. A general compound distribution of this bivariate gamma extension is investigated in the present paper. Properties of this compound distribution are studied, and a compound bivariate gamma-beta distribution and a compound bivariate gamma-exponential distribution are presented as important examples.

1. INTRODUCTION

A compound distribution arises if one or more of the parameters of a given distribution, called the parent distribution, are viewed as random variables. In the continuous case, which will be considered in this paper, the probability density function (PDF) of the compound distribution is obtained by taking the expectation of the parent PDF with respect to the distribution assumed for the random parameter(s). This latter distri-

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bution is termed the mixing distribution.

Mixing is a useful mechanism for constructing new distributions which are not only of theoretical interest but also of practical value. Sichel (1957) is an early reference that provides an application of a univariate compound exponential-gamma distribution to describe the size of airborne mine dust. Later, Bhattacharya (1966) found the same distribution useful to describe a reliability situation with a high percentage of early failures.

Compound distributions of the univariate gamma distribution were discussed in Dubey (1970), while Nagarsenker (1970) considered compound distributions of the generalised gamma distribution introduced in Stacy (1962). In the bivariate case, Hutchinson (1981) studied compound distributions of the distribution consisting of the product of two independent univariate gamma distributions. Roux and Becker (1981) considered compound distributions of the bivariate gamma extension introduced in Becker and Roux (1981), and found that these distributions could be used to describe bivariate processes with a high percentage of early failures.

Steel and Le Roux (1987) suggested a reparameterisation of the bivariate gamma extension introduced by Becker and Roux (1981). This reparameterisation offers the advantage that simple parameter restrictions suffice to identify models for various important physical situations. The present paper not only introduces a class of compound distributions of this reparameterised bivariate gamma extension, but also illustrates the way in which the nature of the situation being modelled, influences the choice of a mixing distribution.

The reparameterised bivariate gamma extension is based on the following model. Consider two components, C_1 and C_2 , operating in a system. These components are subject to shocks, S_1 and S_2 respectively. It is assumed that C_1 fails after receiving h shocks and C_2 fails after receiving ℓ shocks. While both components are functioning, the occurrence of S_j is governed by a Poisson process with parameter $\frac{1}{\alpha_j}$, $j = 1, 2$. If C_1 fails first, a parameter shift occurs and the subsequent occurrence of S_2 is governed by a Poisson process with parameter $\frac{\lambda_2}{\alpha_2}$. A similar parameter

shift, $\frac{1}{\alpha_1} \rightarrow \frac{\lambda_1}{\alpha_1}$, takes place if C_2 fails first. The four possible Poisson processes governing the occurrence of shocks are assumed independent.

In the above model simple restrictions on the values of λ_1 and λ_2 suffice to identify models for various plausible physical situations. If $0 < \lambda_1, \lambda_2 < 1$, a competition model arises: the components (or more realistically for this case, the individuals) compete for the same limited resources, and failure (or death) of one leads to a decrease in the rate at which shocks subsequently strike the remaining component (or individual).

If $\lambda_1, \lambda_2 > 1$, a shared load model is obtained: C_1 and C_2 share a certain load, and failure of one leads to an increase in the rate at which shocks subsequently strike the remaining component. An example of such a situation is the functioning of the two kidneys in the human body.

A third possibility arises when $\lambda_1 > 1$ and $0 < \lambda_2 < 1$, or vice versa. Such an asymmetrical model would be appropriate, for instance, if C_1 represents a parasite with C_2 as host.

Finally, if $\lambda_1 = \lambda_2 = 1$, no parameter shift occurs, and the components act independently.

The bivariate density of the continuous random variables X_1 and X_2 , indicating the respective lifetimes of C_1 and C_2 , was derived in Steel and Le Roux (1987) viz.

(1.1)

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} \frac{\lambda_2 x_1^{h-1}}{\Gamma(h)\Gamma(\ell)\alpha_1^h \alpha_2^\ell} [\lambda_2(x_2 - x_1) + x_1]^{\ell-1} \\ \cdot \exp \left[- \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} - \frac{\lambda_2}{\alpha_2} \right) x_1 - \frac{\lambda_2}{\alpha_2} x_2 \right], \\ \quad 0 < x_1 < x_2 \\ \frac{\lambda_1 x_2^{\ell-1}}{\Gamma(h)\Gamma(\ell)\alpha_1^h \alpha_2^\ell} [\lambda_1(x_1 - x_2) + x_2]^{h-1} \\ \cdot \exp \left[- \frac{\lambda_1}{\alpha_1} x_1 - \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} - \frac{\lambda_1}{\alpha_1} \right) x_2 \right], \\ \quad 0 < x_2 < x_1. \end{cases}$$

In the special case $h = \ell = 1$, the density (1.1) becomes

(1.2)

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} \frac{\lambda_2}{\alpha_1 \alpha_2} \exp \left[- \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} - \frac{\lambda_2}{\alpha_2} \right) x_1 - \frac{\lambda_2}{\alpha_2} x_2 \right], & 0 < x_1 < x_2 \\ \frac{\lambda_1}{\alpha_1 \alpha_2} \exp \left[- \frac{\lambda_1}{\alpha_1} x_1 - \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} - \frac{\lambda_1}{\alpha_1} \right) x_2 \right], & 0 < x_2 < x_1. \end{cases}$$

This is the PDF of a reparameterisation of the bivariate exponential extension introduced by Freund (1961). Weier (1981) utilised a special case of (1.2) to obtain Bayesian estimates of the Freund parameters. The results obtained in the present paper generalise Weier's model to the case where $h, \ell > 1$, yielding a bivariate gamma model.

When studying compound distributions of (1.1), it is convenient to transform to random variables $Y_1 = \text{minimum}(X_1, X_2)$ and $Y_2 = \text{maximum}(X_1, X_2) - \text{minimum}(X_1, X_2)$. This transformation applied to (1.1) gave the joint PDF of Y_1 and Y_2 as

$$(1.3) \quad f_{Y_1, Y_2}(y_1, y_2) = \frac{\exp \left[- \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} \right) y_1 \right]}{\Gamma(h) \Gamma(\ell) \alpha_1^h \alpha_2^\ell} \left[\lambda_1 y_1^{\ell-1} (y_1 + \lambda_1 y_2)^{h-1} \right. \\ \left. \exp \left(- \frac{\lambda_1}{\alpha_1} y_2 \right) + \lambda_2 y_1^{h-1} (y_1 + \lambda_2 y_2)^{\ell-1} \right. \\ \left. \exp \left(- \frac{\lambda_2}{\alpha_2} y_2 \right) \right], y_1, y_2 > 0.$$

2. A GENERAL COMPOUND DISTRIBUTION OF THE REPARAMETERISED BIVARIATE GAMMA EXTENSION

Compound distributions of the bivariate gamma extension with PDF (1.3) are now considered by viewing λ_1 and λ_2 as values of random variables, Λ_1 and Λ_2 , with PDF's $f_{\Lambda_1}(\lambda_1)$ and $f_{\Lambda_2}(\lambda_2)$ respectively. It is assumed that Λ_1 and Λ_2 are independently distributed, and that $f_{\Lambda_1}(\lambda_1) > 0$ for $\lambda_1 \in A_1$, $f_{\Lambda_2}(\lambda_2) > 0$ for $\lambda_2 \in A_2$, where A_1 and A_2 are possibly infinite subintervals of $(0, \infty)$.

The next theorem provides an expression for the compound PDF of Y_1 and Y_2 .

Theorem 2.1 The joint compound (or unconditional) PDF of Y_1 and Y_2 is

$$\begin{aligned}
 (2.1) \quad & f_{Y_1, Y_2}(y_1, y_2) \\
 &= \frac{\exp\left[-\left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2}\right)y_1\right] y_1^{h+\ell-2}}{\Gamma(h)\Gamma(\ell)\alpha_1^h\alpha_2^\ell} \left[\sum_{k=0}^{h-1} \binom{h-1}{k} \left(\frac{y_2}{y_1}\right)^k \int_{A_1} \lambda_1^{k+1} \right. \\
 &\quad \exp\left(-\frac{y_2}{\alpha_1}\lambda_1\right) f_{\Lambda_1}(\lambda_1) d\lambda_1 + \sum_{k=0}^{\ell-1} \binom{\ell-1}{k} \left(\frac{y_2}{y_1}\right)^k \int_{A_2} \lambda_2^{k+1} \\
 &\quad \left. \exp\left(-\frac{y_2}{\alpha_2}\lambda_2\right) f_{\Lambda_2}(\lambda_2) d\lambda_2 \right], y_1, y_2 > 0.
 \end{aligned}$$

Proof In order to obtain (2.1), apply the binomial theorem to (1.3), and take the expectation of the resulting expression with respect to $f_{\Lambda_1}(\lambda_1)$ and $f_{\Lambda_2}(\lambda_2)$.

The following theorem summarises some of the properties of the compound distribution with PDF (2.1).

Theorem 2.2

(i) The joint survival distribution function of Y_1 and Y_2 is

$$\begin{aligned}
 (2.2) \quad & \bar{F}_{Y_1, Y_2}(s, t) \equiv P(Y_1 > s, Y_2 > t) \\
 &= \frac{\alpha_1^t \alpha_2^h}{\Gamma(h)\Gamma(\ell)} \left[\sum_{k=0}^{h-1} \binom{h-1}{k} \frac{\Gamma\left(h+\ell-k-1; \frac{\alpha_1+\alpha_2}{\alpha_1\alpha_2}s\right)}{\alpha_2^{k+1}(\alpha_1+\alpha_2)^{h+\ell-k-1}} \right. \\
 &\quad \cdot \int_{A_1} f_{\Lambda_1}(\lambda_1) \Gamma\left(k+1; \frac{t\lambda_1}{\alpha_1}\right) d\lambda_1 + \sum_{k=0}^{\ell-1} \binom{\ell-1}{k} \\
 &\quad \cdot \frac{\Gamma\left(h+\ell-k-1; \frac{\alpha_1+\alpha_2}{\alpha_1\alpha_2}s\right)}{\alpha_1^{k+1}(\alpha_1+\alpha_2)^{h+\ell-k-1}} \int_{A_2} f_{\Lambda_2}(\lambda_2) \Gamma\left(k+1; \frac{t\lambda_2}{\alpha_2}\right) d\lambda_2 \left. \right], \\
 &s, t > 0,
 \end{aligned}$$

where $\Gamma(k; z) = \int_z^\infty \exp(-u) u^{k-1} du$, $z > 0$, is related to the incomplete gamma function.

(ii) The marginal PDF of Y_1 is given by

$$(2.3) \quad f_{Y_1}(y_1) = \frac{\exp \left[- \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} \right) y_1 \right] y_1^{h+\ell-2}}{\Gamma(h)\Gamma(\ell)\alpha_1^h\alpha_2^\ell} \\ \left[\alpha_1 \sum_{k=0}^{h-1} \binom{h-1}{k} \Gamma(k+1) \left(\frac{\alpha_1}{y_1} \right)^k \right. \\ \left. + \alpha_2 \sum_{k=0}^{\ell-1} \binom{\ell-1}{k} \Gamma(k+1) \left(\frac{\alpha_2}{y_1} \right)^k \right], y_1 > 0,$$

while that of Y_2 is given by

$$(2.4) \quad f_{Y_2}(y_2) = \frac{\alpha_1^{\ell-1}\alpha_2^{h-1}}{\Gamma(h)\Gamma(\ell)(\alpha_1 + \alpha_2)^{h+\ell-1}} \\ \left[\sum_{k=0}^{h-1} \binom{h-1}{k} \left(\frac{\alpha_1 + \alpha_2}{\alpha_1\alpha_2} y_2 \right)^k \Gamma(h + \ell - k - 1) \right. \\ \left. \int_{A_1} \lambda_1^{k+1} \exp \left(- \frac{y_2}{\alpha_1} \lambda_1 \right) f_{\Lambda_1}(\lambda_1) d\lambda_1 + \sum_{k=0}^{\ell-1} \binom{\ell-1}{k} \left(\frac{\alpha_1 + \alpha_2}{\alpha_1\alpha_2} y_2 \right)^k \right. \\ \left. \Gamma(h + \ell - k - 1) \int_{A_2} \lambda_2^{k+1} \exp \left(- \frac{y_2}{\alpha_2} \lambda_2 \right) f_{\Lambda_2}(\lambda_2) d\lambda_2 \right], y_2 > 0.$$

(iii) The joint moments of Y_1 and Y_2 can be found from

$$(2.5) \quad \mu'_{rs} \equiv E(Y_1^r Y_2^s) \\ = \frac{\alpha_1^{\ell+r}\alpha_2^{h+r}}{\Gamma(h)\Gamma(\ell)(\alpha_1 + \alpha_2)^{h+\ell+r}} \\ \left[\alpha_1^s \sum_{k=0}^{h-1} \binom{h-1}{k} \Gamma(h + \ell + r - k - 1) \right. \\ \Gamma(k + s + 1) \left(\frac{\alpha_1 + \alpha_2}{\alpha_2} \right)^{k+1} \int_{A_1} \lambda_1^{-s} f_{\Lambda_1}(\lambda_1) d\lambda_1 + \alpha_2^s \sum_{k=0}^{\ell-1} \\ \left(\frac{\ell-1}{k} \right) \Gamma(h + \ell + r - k - 1) \Gamma(k + s + 1) \left(\frac{\alpha_1 + \alpha_2}{\alpha_1} \right)^{k+1} \\ \left. \int_{A_2} \lambda_2^{-s} f_{\Lambda_2}(\lambda_2) d\lambda_2 \right], r, s = 0, 1, 2, \dots$$

Proof The results follow in the usual manner from the bivariate PDF (2.1).

Remark 2.1

(i) If $h = \ell = 1$, (2.2) becomes

$$(2.6) \quad \bar{F}_{Y_1, Y_2}(s, t) = \frac{1}{\alpha_1 + \alpha_2} \exp\left(-\frac{\alpha_1 + \alpha_2}{\alpha_1 \alpha_2} s\right) \left[\alpha_1 \int_{A_1} f_{\Lambda_1}(\lambda_1) \exp\left(-\frac{t}{\alpha_1} \lambda_1\right) d\lambda_1 + \alpha_2 \int_{A_2} f_{\Lambda_2}(\lambda_2) \exp\left(-\frac{t}{\alpha_2} \lambda_2\right) d\lambda_2 \right], s, t > 0.$$

It is evident that in this special case Y_1 and Y_2 are independently distributed. The distribution of Y_1 is exponential $\left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2}\right)$, while that of Y_2 is a (finite) mixture of two distributions, with mixing coefficients $\frac{\alpha_1}{\alpha_1 + \alpha_2}$ and $\frac{\alpha_2}{\alpha_1 + \alpha_2}$. These latter two distributions are obtained by mixing exponential $\left(\frac{\lambda_j}{\alpha_j}\right)$ distributions with prior distributions for $\Lambda_j, j = 1, 2$.

(ii) The marginal PDF (2.3) is a finite mixture of gamma PDF's while (2.4) depends on the specific mixing distributions assumed for Λ_1 and Λ_2 .

(iii) It is clear from (2.5) that $\mu'_{r,s}$ exists only if $E(\Lambda_1^{-s})$ and $E(\Lambda_2^{-s})$ exist. For example, if Λ_1 and Λ_2 have beta type I distributions with parameters a and b , $E(\Lambda_j^{-s})$ exist only if $s < a$. Hence, if these mixing distributions are assumed, the moments, $\mu'_{r,s}$, of the resulting compound distribution exist only if $s < a$.

3. TWO IMPORTANT EXAMPLES

It was previously mentioned that in (1.3), $0 < \lambda_1, \lambda_2 < 1$ identifies a competition model, while $\lambda_1, \lambda_2 > 1$ identifies a shared load model. An important advantage of the reparameterised bivariate gamma extension now becomes clear: the nature of the practical situation being modelled determines a class of suitable mixing or Bayes prior distributions. If a compound distribution for a competition model is required, mixing distributions on $(0,1)$ are appropriate, while mixing distributions on $(1, \infty)$ yield compound distributions for a shared load model. The character of the

practical situation being modelled thus determines a class of appropriate mixing distributions. If Bayesian estimates of λ_1 and λ_2 are required, the above considerations similarly suggest appropriate prior distributions for Λ_1 and Λ_2 .

Now suppose that a compound distribution for a competition model is required, implying that the mixing distributions in (2.1) should be defined on the interval (0,1). Assume that these distributions are in fact identical beta type I distributions, with parameters a and b , i.e.

$$(3.1) \quad f_{\Lambda_j}(\lambda_j) = \frac{1}{B(a,b)} \lambda_j^{a-1} (1-\lambda_j)^{b-1}, \quad 0 < \lambda_j < 1,$$

for $j = 1, 2$. From (2.1),

$$\begin{aligned} & f_{Y_1, Y_2}(y_1, y_2) \\ &= \frac{\exp \left[- \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} \right) y_1 \right] y_1^{h+\ell-2}}{\Gamma(h)\Gamma(\ell)\alpha_1^h \alpha_2^\ell B(a,b)} \cdot \\ & \quad \left[\sum_{k=0}^{h-1} \binom{h-1}{k} \left(\frac{y_2}{y_1} \right)^k \int_0^1 \lambda_1^{a+k} (1-\lambda_1)^{b-1} \exp \left(-\frac{y_2}{\alpha_1} \lambda_1 \right) d\lambda_1 \right. \\ & \quad \left. + \sum_{k=0}^{\ell-1} \binom{\ell-1}{k} \left(\frac{y_2}{y_1} \right)^k \int_0^1 \lambda_2^{a+k} (1-\lambda_2)^{b-1} \exp \left(-\frac{y_2}{\alpha_2} \lambda_2 \right) d\lambda_2 \right]. \end{aligned}$$

The integrals in the above expression can be solved by using an integral representation (cf. Luke, 1969, p. 115) of the confluent hypergeometric function,

$${}_1F_1(c; d; z) = \sum_{n=0}^{\infty} \frac{(c)_n z^n}{(d)_n n!}.$$

This yields

$$(3.2) \quad f_{Y_1, Y_2}(y_1, y_2)$$

$$= \frac{\Gamma(a+b) \exp \left[- \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} \right) y_1 \right] y_1^{h+\ell-2}}{\Gamma(h) \Gamma(\ell) \Gamma(a) \alpha_1^h \alpha_2^\ell} \cdot$$

$$\left[\sum_{k=0}^{h-1} \binom{h-1}{k} \left(\frac{y_2}{y_1} \right)^k \frac{\Gamma(a+k+1)}{\Gamma(a+b+k+1)} {}_1F_1(a+k+1; a+b+k+1; \right.$$

$$\left. - \frac{y_2}{\alpha_1} \right) + \sum_{k=0}^{\ell-1} \binom{\ell-1}{k} \left(\frac{y_2}{y_1} \right)^k \frac{\Gamma(a+k+1)}{\Gamma(a+b+k+1)} {}_1F_1(a+k+1; a+b+k+1; -\frac{y_2}{\alpha_2}) \right].$$

Expression (3.2) is the PDF of a compound bivariate gamma-beta distribution. Properties of this distribution follow from theorem 2.2 by appropriate substitution of (3.1).

Also, if a compound distribution for a shared load model is required, the mixing distributions in (2.1) should be defined on the interval $(1, \infty)$. Assume that these distributions are in fact identical exponential distributions, i.e.

$$(3.3) \quad f_{\Lambda_j}(\lambda_j) = \theta \exp[-\theta(\lambda_j - 1)], \quad 1 < \lambda_j < \infty,$$

for $j = 1, 2$. From (2.1),

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{\theta \exp \left[- \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} \right) y_1 \right] y_1^{h+\ell-2}}{\Gamma(h) \Gamma(\ell) \alpha_1^h \alpha_2^\ell} \left[\sum_{k=0}^{h-1} \binom{h-1}{k} \left(\frac{y_2}{y_1} \right)^k \right.$$

$$\cdot \int_1^\infty \lambda_1^{k+1} \exp \left[-\frac{y_2}{\alpha_1} \lambda_1 - \theta(\lambda_1 - 1) \right] d\lambda_1 + \sum_{k=0}^{\ell-1} \binom{\ell-1}{k} \left(\frac{y_2}{y_1} \right)^k$$

$$\cdot \int_1^\infty \lambda_2^{k+1} \exp \left[-\frac{y_2}{\alpha_2} \lambda_2 - \theta(\lambda_2 - 1) \right] d\lambda_2 \Big].$$

Transforming $u = \lambda_1 - 1$, $v = \lambda_2 - 1$, and applying the binomial theorem leads to

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{\theta \exp \left[- \left(\frac{1}{\alpha_1} + \frac{1}{\alpha_2} \right) y_1 \right] y_1^{h+\ell-2}}{\Gamma(h)\Gamma(\ell)\alpha_1^h\alpha_2^\ell} \left[\exp \left(-\frac{y_2}{\alpha_1} \right) \sum_{k=0}^{h-1} \binom{h-1}{k} \left(\frac{y_2}{y_1} \right)^k \sum_{j=0}^{k+1} \binom{k+1}{j} \Gamma(j+1) \left(\theta + \frac{y_2}{\alpha_1} \right)^{-(j+1)} + \exp \left(-\frac{y_2}{\alpha_2} \right) \sum_{k=0}^{\ell-1} \binom{\ell-1}{k} \left(\frac{y_2}{y_1} \right)^k \sum_{j=0}^{k+1} \binom{k+1}{j} \Gamma(j+1) \left(\theta + \frac{y_2}{\alpha_2} \right)^{-(j+1)} \right],$$

$y_1, y_2 > 0.$

The properties of this compound bivariate gamma-exponential distribution are obtained if (3.3) is appropriately substituted in theorem 2.2.

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